

## Seminar Nr. 5, Probability and Statistics

### Continuous Random Variables and Continuous Random Vectors

#### Theory Review

$X : S \rightarrow \mathbb{R}$  continuous random variable with pdf  $f : \mathbb{R} \rightarrow \mathbb{R}$  and cdf  $F : \mathbb{R} \rightarrow \mathbb{R}$ . Properties:

$$1. F(x) = P(X \leq x) = \int_{-\infty}^x f(t)dt$$

$$2. f(x) \geq 0, \forall x \in \mathbb{R}, \int_{\mathbb{R}} f(x) = 1$$

$$3. P(X = x) = 0, \forall x \in \mathbb{R}, P(a < X < b) = P(a \leq X \leq b) = \int_a^b f(t)dt$$

$$4. F(-\infty) = 0, F(\infty) = 1$$

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$(X, Y) : S \rightarrow \mathbb{R}^2$  continuous random vector with pdf  $f = f_{(X,Y)} : \mathbb{R}^2 \rightarrow \mathbb{R}$  and

cdf  $F = F_{(X,Y)} : \mathbb{R}^2 \rightarrow \mathbb{R}$ ,  $F(x, y) = P(X \leq x, Y \leq y) = \int_{-\infty}^x \int_{-\infty}^y f(u, v) dv du, \forall (x, y) \in \mathbb{R}^2$ . Properties:

$$1. P(a_1 < X \leq b_1, a_2 < Y \leq b_2) = F(b_1, b_2) - F(a_1, b_2) - F(b_1, a_2) + F(a_1, a_2)$$

$$2. F(\infty, \infty) = 1, F(-\infty, y) = F(x, -\infty) = 0, \forall x, y \in \mathbb{R}$$

$$3. F_X(x) = F(x, \infty), F_Y(y) = F(\infty, y), \forall x, y \in \mathbb{R} \text{ (marginal cdf's)}$$

$$4. P((X, Y) \in D) = \int_D \int f(x, y) dy dx$$

$$5. f_X(x) = \int_{\mathbb{R}} f(x, y) dy, \forall x \in \mathbb{R}, f_Y(y) = \int_{\mathbb{R}} f(x, y) dx, \forall y \in \mathbb{R} \text{ (marginal densities)}$$

$$6. X \text{ and } Y \text{ are independent} \Leftrightarrow f_{(X,Y)}(x, y) = f_X(x)f_Y(y), \forall (x, y) \in \mathbb{R}^2.$$

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**Function**  $Y = g(X)$ :  $X$  r.v.,  $g : \mathbb{R} \rightarrow \mathbb{R}$  differentiable with  $g' \neq 0$ , strictly monotone

$$f_Y(y) = \frac{f_X(g^{-1}(y))}{|g'(g^{-1}(y))|}, y \in g(\mathbb{R})$$

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**Uniform distribution**  $U(a, b)$ ,  $-\infty < a < b < \infty$ : pdf  $f(x) = \frac{1}{b-a}, x \in [a, b]$ .

**Normal distribution**  $N(\mu, \sigma)$ ,  $\mu \in \mathbb{R}, \sigma > 0$ : pdf  $f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, x \in \mathbb{R}$ .

**Gamma distribution**  $Gamma(a, b)$ ,  $a, b > 0$ : pdf  $f(x) = \frac{1}{\Gamma(a)b^a} x^{a-1} e^{-\frac{x}{b}}, x > 0$ .

**Exponential distribution**  $Exp(\lambda) = Gamma(1, 1/\lambda)$ ,  $\lambda > 0$ : pdf  $f(x) = \lambda e^{-\lambda x}, x > 0$ .

- Exponential distribution models *time*: waiting time, interarrival time, failure time, time between rare events, etc; the parameter  $\lambda$  represents the frequency of rare events, measured in  $\text{time}^{-1}$ .

- Gamma distribution models the *total* time of a multistage scheme.

- For  $\alpha \in \mathbb{N}$ , a  $Gamma(\alpha, 1/\lambda)$  variable is the sum of  $\alpha$  independent  $Exp(\lambda)$  variables.

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1. The lifetime, in years, of some electronic component is a random variable with density

$$f(x) = \begin{cases} \frac{k}{x^4}, & \text{for } x \geq 1 \\ 0, & \text{for } x < 1. \end{cases}$$

Find

- a) the constant  $k$ ;
- b) the corresponding cdf  $F$ ;
- c) the probability for the lifetime of the component to exceed 2 years.

2. (The Uniform property) Let  $X \in U(a, b)$ . For any  $h > 0$  and  $t, s \in [a, b - h]$ ,

$$P(s < X < s + h) = P(t < X < t + h).$$

The probability is only determined by the length of the interval, but not by its location.

Example: A certain flight can arrive at any time between 4:50 and 5:10 pm. Let  $X$  denote the arrival time of the flight.

- a) What distribution does  $X$  have?
- b) When is the flight more likely to arrive: between 4:50 and 4:55 or between 5 and 5:05; before 5 or after 5?

3. On the average, a computer experiences breakdowns every 5 months. The time until the first breakdown and the times between any two consecutive breakdowns are independent Exponential random variables. After the third breakdown, a computer requires a special maintenance.

- a) Find the probability that a special maintenance is required within the next 9 months;
- b) Given that a special maintenance was not required during the first 12 months, what is the probability that it will not be required within the next 4 months?

4. The joint density for  $(X, Y)$  is  $f_{(X,Y)}(x, y) = \frac{1}{16}x^3y^3, x, y \in [0, 2]$ .

- a) Find the marginal densities  $f_X, f_Y$ .
- b) Are  $X$  and  $Y$  independent?
- c) Find  $P(X \leq 1)$ .

5. Let  $X$  be a random variable with density  $f_X(x) = \frac{1}{4}xe^{-\frac{x}{2}}, x \geq 0$  and let  $Y = \frac{1}{2}X + 2$ . Find  $f_Y$ .

6. Let  $X \in N(0, 1)$ . Find the probability density function of  $Y = |X|$ .

### Bonus Problems:

7. Let  $X$  denote the velocity of a random gas molecule. According to the Maxwell-Boltzmann law, the pdf of  $X$  is given by

$$f_X(x) = cx^2e^{-\beta x^2}, x > 0,$$

where the constants  $c$  and  $\beta$  depend on the gas involved, its mass and its temperature. The kinetic energy of the molecule is given by  $Y = \frac{1}{2}mX^2$ , where  $m > 0$ . For a gas molecule with  $c = 2, \beta = 1$  and  $m = 1$ , find the pdf of the kinetic energy of the molecule.

8. A gamer shoots at a virtual shooting board centered at the origin of the Cartesian coordinate system such that the coordinates of the hit are two independent random variables that follow the  $N(0, 1)$  distribution. Find the probability that the shooter hits the upper half-plane of the shooting board at a distance between 1 and 2 from the origin.